

Question

Write a Java program to demonstrate multiple inheritance using interfaces.

Steps for Execution

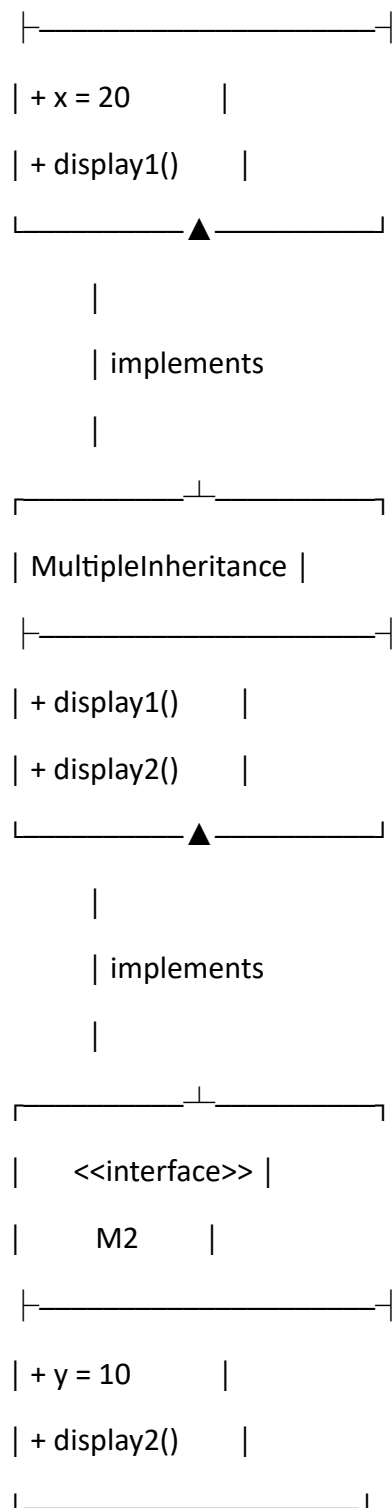
1. Create interface M1.
2. Declare constant x = 20 and method display1() in M1.
3. Create interface M2.
4. Declare constant y = 10 and method display2() in M2.
5. Create class MultipleInheritance implementing both M1 and M2.
6. Override display1() and display2() in the class.
7. Create an object of MultipleInheritance.
8. Access the variable x using M1.x.
9. Call display1() using the object.
10. Compile and execute the program.

Algorithm

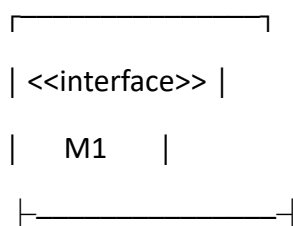
1. **Start**
2. Define interface M1 with constant x = 20 and method display1().
3. Define interface M2 with constant y = 10 and method display2().
4. Define class MultipleInheritance implementing M1 and M2.
5. Implement display1() to print "hello".
6. Implement display2() to print "bye".
7. Create an object m1 of MultipleInheritance.
8. Display the value of M1.x.
9. Call display1().
10. **Stop**

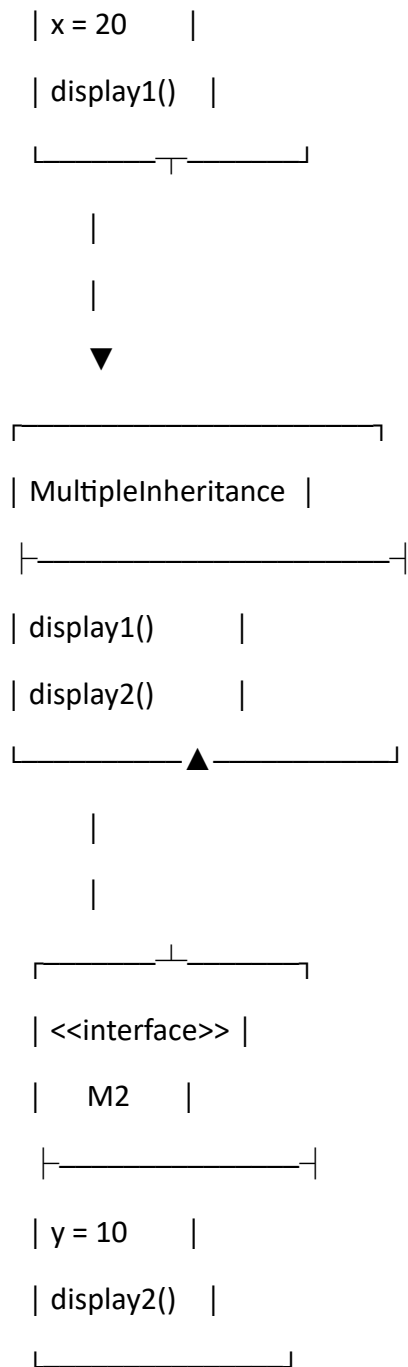
UML Class Diagram





A clearer UML representation is:





Code

```
package javaprograms;
```

```
interface M1 {
```

```
    int x = 20;
```

```
    void display1();
```

```
}
```

```
interface M2 {
```

```
    int y = 10;
```

```
    void display2();
```

```
}
```

```
public class MultipleInheritance implements M1, M2 {
```

```
    @Override
```

```
    public void display1() {
```

```
        System.out.println("hello");
```

```
    }
```

```
    @Override
```

```
    public void display2() {
```

```
        System.out.println("bye");
```

```
    }
```

```
    public static void main(String[] args) {
```

```
        MultipleInheritance m1 = new MultipleInheritance();
```

```
        System.out.println("result is: " + M1.x);
```

```
        m1.display1();
```

```
    }
```

```
}
```

Output

result is: 20

hello